

PROBLEM/OPPORTUNITY STATEMENT

Table 1: Registration and Data Management

Brief Statement of Problem, Opportunity or Directive

The university currently manages all postgraduate student data using a single, massive spreadsheet. This spreadsheet tracks every detail from registration through to final examination, making it overwhelming and difficult to navigate for administrative staff.

Cause and Effect

Cause: The system grew organically over time without proper planning. As student numbers increased, more columns and sheets were added to accommodate new data requirements rather than implementing a proper database solution.

Effect: Administrative staff spend excessive time scrolling through thousands of rows to find specific student records. Data entry errors are common due to manual copying and pasting. Information is often duplicated across different sections of the spreadsheet, leading to inconsistencies.

System Objective

To implement a centralized database system that stores student information in a structured, normalized format. The system will provide role-based views so that staff only see relevant data for their tasks. Search and filter functions will allow users to find specific records within seconds rather than minutes.

System Constraint

The system must comply with data protection regulations (POPIA/GDPR). It must be web-based to allow access from multiple campuses. Development must be completed within the current academic year budget. Legacy data from the existing spreadsheet must be migratable without loss.

Problem/Opportunity Statement Table 2: Milestone and Deadline Tracking

Brief Statement of Problem, Opportunity or Directive

Postgraduate students have strict academic milestones (6-month reports for Masters, 1-year reports for PhD), but there is no automated system to track these deadlines or notify students and supervisors when deadlines are approaching or have been missed.

Cause and Effect

Cause: The current spreadsheet is static and cannot generate automatic reminders. Responsibility for tracking deadlines falls on individual supervisors and administrative staff, who must manually check dates against the current calendar.

Effect: Students frequently miss submission deadlines, delaying their academic progress. Supervisors forget to submit progress reports. Administrative staff must spend hours each month manually generating reminder emails. Some students have fallen behind by months before anyone notices.

System Objective

To implement an automated notification engine that tracks all student milestones and deadlines. The system will automatically send email reminders to students, supervisors, and administrators at configured intervals before each deadline. A dashboard will show all upcoming and overdue milestones in real-time.

System Constraint

The system must allow flexible configuration of deadline rules for different programs and cohorts. Notifications must be trackable to ensure they were sent. The system must handle timezone differences for international students and external examiners.

Problem/Opportunity Statement Table 3: Proposal Review and Feedback Process

Brief Statement of Problem, Opportunity or Directive

The process of submitting research proposals, assigning them to evaluators, collecting feedback, and communicating decisions to students is disjointed and relies heavily on email, leading to lost documents, delayed feedback, and lack of transparency.

Cause and Effect

Cause: There is no centralized platform for proposal submission and review. Students email proposals to administrative staff, who then forward them to evaluators via email. Evaluators submit feedback via email reply, which must be manually collated and forwarded back to students.

Effect: Proposals are sometimes lost in email inboxes. Evaluators forget to complete reviews without reminders. Students have no visibility into where their proposal is in the review process and constantly email staff asking for status updates. Feedback is inconsistent in format and difficult to compare across evaluators.

System Objective

To create a structured workflow for proposal submission and review. Students will upload proposals directly to the system. Administrators will assign evaluators through the interface. Evaluators will receive automated notifications and will submit

feedback through standardized forms. Students will see real-time status updates (Submitted, Under Review, Feedback Sent) on their dashboard.

System Constraint

The system must support multiple file formats for uploads (PDF, DOCX). It must maintain a complete audit trail of who accessed each proposal and when. Feedback forms must be customizable by faculty. The system must handle confidential feedback that is not visible to students until released by administrators.

Problem/Opportunity Statement Table 4: Reporting and Analytics

Brief Statement of Problem, Opportunity or Directive

Administrators and the SPGS Chair cannot easily generate reports on student progress, proposal outcomes, ethics approvals, or performance metrics because all data is trapped in an unstructured spreadsheet that requires manual manipulation to extract insights.

Cause and Effect

Cause: The spreadsheet was designed for data entry, not analysis. There are no built-in reporting tools, and the unstructured nature of the data makes it difficult to run consistent queries.

Effect: When reporting is required for university management or accreditation bodies, staff must spend days manually extracting, cleaning, and formatting data. Reports are often inconsistent because different staff members extract data differently. Strategic decisions are made based on incomplete or outdated information.

System Objective

To implement an integrated reporting tool that allows administrators to generate real-time reports on any aspect of the postgraduate lifecycle. Predefined reports will cover proposal outcomes, ethics application statuses, student progression rates, and supervisor performance metrics. Reports will be exportable to common formats (PDF, Excel).

System Constraint

Reporting must respect data privacy rules—only authorized users can see certain data. Reports must be auditable to track who generated them and when. The system must support ad-hoc querying while maintaining performance. Historical data must be preserved for longitudinal analysis.

The PIECES Framework

Problem/ Opportunity Area	Related problems/ opportunities identified (yes/no)	Specific problems/opportunities identified within the area
Performance		Speed: Loading and navigating the massive spreadsheet takes 30-60 seconds even on modern computers. Scrolling through 5,000+ rows is slow and cumbersome. Accuracy: Manual data entry leads to typos, inconsistent formatting, and duplicate records. When multiple staff edit the same file, changes are overwritten.
Information (and Data)		Role-based dashboards will show each user only the information relevant to their tasks. Standardized forms will ensure all required data is captured. Students will have a personal progress portal showing exactly where they stand.
Economics		Automation will reduce manual data management time by an estimated 80%. Automated notifications will eliminate hours of manual email drafting. Self-service student portals will reduce ad-hoc queries to staff.
Control (and security)		Access control: The spreadsheet is stored on a shared drive with folder-level permissions. Anyone with access can see all student data, including

		sensitive information. Audit trail: There is no record of who viewed or changed data. If a student's record is altered incorrectly, there is no way to trace who did it or restore the previous version. Data loss: No automated backups. If the file becomes corrupted, significant data could be lost.
Efficiency		Automated workflows will route proposals directly from submission to assignment to review without manual email handling. Data entered once (e.g., student registration) will automatically populate all relevant modules, eliminating redundant data entry.
Services		User satisfaction: Students frequently complain about lack of visibility into their progress. Supervisors find it difficult to track their students' milestones. Evaluators dislike the manual email-based review process. Support burden: Staff spend hours responding to "Where is my proposal?" and "When is my deadline?" queries.

Project Scope

Information Scope (Business Data Requirements)

Requirements are prioritised using MoSCoW: (M)ust Have, (S)hould Have, (C)ould Have, (W)on't Have

ID	Entity	Description	Priority
INF001	Student	Personal details (name, ID number, contact information, student number, program of study, registration date)	M
INF002	User	System user details (username, password hash, role, last login, account status)	M
INF003	Role	User role definitions (Student, Supervisor, Administrator, Evaluator) with associated permissions	M
INF004	Supervisor Assignment	Relationship between students and supervisors (primary supervisor, co-supervisor, assignment date, status)	M
INF005	Proposal	Research proposal submissions (title, abstract, document file, submission date, current status, version number)	M
INF006	Proposal Review	Evaluator feedback on proposals (evaluator ID, proposal ID, feedback text, score/rating, submission date, confidential notes)	M

ID	Entity	Description	Priority
INF007	Ethics Application	Ethics clearance applications (application details, supporting documents, submission date, status, approval date)	M
INF008	Milestone	Predefined milestone templates (name, description, default deadline offset, program type)	M
INF009	Student Milestone	Student-specific milestone instances (milestone ID, student ID, due date, completion date, status, supporting document)	M
INF010	Notification	Notification records (recipient, type, subject, message, sent date, read status)	M
INF011	Document	Generic document storage (filename, file path, uploader ID, upload date, associated record ID, document type)	M
INF012	Program	Program of study definitions (program name, code, duration, default milestones)	S
INF013	Faculty	Faculty/Department structure (faculty name, code, associated programs)	S
INF014	Audit Log	System audit trail (user ID, action, timestamp, IP address, affected record type, record ID, old value, new value)	S
INF015	Feedback Template	Configurable templates for evaluator feedback forms (questions, rating scales, required fields)	C
INF016	Report Definition	Saved report configurations (name, parameters, created by, schedule)	C

ID	Entity	Description	Priority
INF017	External Examiner	External examiner details (name, institution, contact information, specialization)	W
INF018	Thesis Submission	Final thesis/dissertation submissions (document, submission date, examiner assignments, examination results)	W

Functional Scope (Business Process Requirements)

ID	Use Case	Description	Priority
UC001	Register Student	Administrative staff register new students in the system, capturing personal details and program information	M
UC002	Login	All users authenticate to access the system	M
UC003	View Dashboard	Users view their personalized dashboard showing relevant information and tasks	M
UC004	Submit Proposal	Students upload and submit their research proposal	M
UC005	Assign Evaluators	Administrators assign evaluators to review submitted proposals	M
UC006	Submit Proposal Feedback	Evaluators submit feedback on assigned proposals	M
UC007	View Proposal Status	Students view the current status of their proposal submission	M
UC008	Submit Ethics Application	Students submit ethics clearance applications	M
UC009	Review Ethics Application	Administrators/Supervisors review and approve/reject ethics applications	M
UC010	Manage Milestones	Administrators create and configure milestone templates	M
UC011	Track Student Milestones	System automatically tracks student progress against milestones	M

ID	Use Case	Description	Priority
UC012	Send Notifications	System sends automated email notifications for deadlines and status changes	M
UC013	View Notifications	Users view their notifications within the system	M
UC014	Generate Report	Administrators generate reports on student progress and outcomes	M
UC015	Manage Users	Administrators create and manage user accounts and roles	M
UC016	Assign Supervisor	Administrators assign supervisors to students	M
UC017	Submit Progress Report	Supervisors submit progress reports for their students	M
UC018	View Student Progress	Supervisors view progress of all their supervised students	S
UC019	Search Students	Staff search for students by various criteria	S
UC020	Export Data	Users export data to PDF or Excel formats	S
UC021	Manage Programs	Administrators create and manage program definitions	S
UC022	View Audit Log	Administrators view system audit trail	S
UC023	Configure Feedback Forms	Administrators customize evaluator feedback templates	C

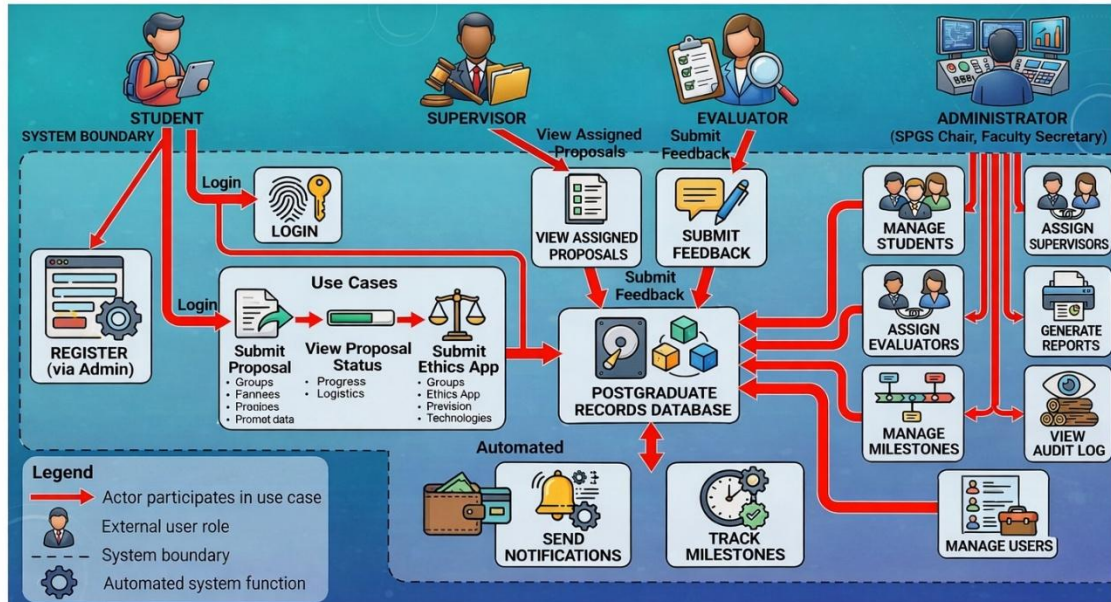
ID	Use Case	Description	Priority
UC024	Schedule Reports	Administrators schedule automatic report generation	C
UC025	Bulk Import Students	Administrators import multiple student records from spreadsheet	C
UC026	Manage External Examiners	Administrators maintain external examiner database	W
UC027	Submit Final Thesis	Students submit final thesis/dissertation	W
UC028	Assign Thesis Examiners	Administrators assign examiners for final thesis	W

Communications Scope (Business & System Interface Requirements)

ID	Interface Requirement	Description	Priority
COM001	Email Notifications	Integration with SMTP email service to send automated notifications to users	M
COM002	Document Upload/Download	File storage and retrieval for proposals, ethics applications, and supporting documents	M
COM003	PDF Export	Export reports and student summaries to PDF format	M
COM004	Excel Export	Export data tables to Microsoft Excel format for further analysis	S
COM005	Web Browser Access	System accessible via standard web browsers (Chrome, Firefox, Edge, Safari)	M
COM006	Responsive Design	Interface adapts to different screen sizes (desktop, tablet, mobile)	S
COM007	Calendar Integration	Export milestones and deadlines to iCal/ICS format for integration with personal calendars	C
COM008	LDAP/Active Directory	Integration with university LDAP for single sign-on authentication	C
COM009	SMS Notifications	Optional SMS notifications for critical deadlines	W

ID	Interface Requirement	Description	Priority
COM010	Student Information System Integration	Integration with existing university SIS to import student data	W
COM011	Cloud Storage Integration	Integration with cloud storage (OneDrive, Google Drive) for document backup	W

High-Level Use Case Diagram



Legend:

→ Actor participates in use case



External user role



System boundary



Automated system function

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POSTGRADUATE RECORD-KEEPING SYSTEM PROJECT OVERVIEW DIAGRAM

